

MID TERM EXAMINATION

Subject: Linked Lists, Stacks, Queues

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 38 | Date: 11-03-2026

Total Questions: 10 | Answer all questions.

Section 1: MCQ (4 Questions - 8 Marks)

Q1. [2 marks] (Easy)

Which of the following is typically stored in each node of a singly linked list, besides the data itself?

Options:

A) Address of the next node B) Address of the previous node C) Size of the list D) Data type of the elements

Q2. [2 marks] (Easy)

What fundamental principle does a Stack data structure follow?

Options:

A) First-In, First-Out (FIFO) B) Last-In, First-Out (LIFO) C) Random Access D) Indexed Access

Q5. [2 marks] (Medium)

In a singly linked list, which operation has a time complexity of $O(1)$ if you have a pointer to the node before the insertion point?

Options:

A) Searching for an element B) Inserting a node at a specific position C) Deleting the last node D) Reversing the list

Q9. [2 marks] (Hard)

What is the worst-case time complexity to find a specific element in an unsorted singly linked list?

Options:

A) $O(1)$ B) $O(\log N)$ C) $O(N)$ D) $O(N \log N)$

Section 2: Short Answer (6 Questions - 30 Marks)

Q3. [5 marks] (Medium)

Differentiate between the PUSH operation in a Stack and the ENQUEUE operation in a Queue, in terms of where the element is added to the data structure.

Q4. [5 marks] (Easy)

State two significant advantages of using Linked Lists over Arrays when dealing with dynamic collections of data.

Q6. [5 marks] (Medium)

Explain how a Stack data structure can be implemented using a singly linked list. Specifically mention how push and pop operations would be performed.

Q7. [5 marks] (Hard)

Describe the process of deleting a specific node from the middle of a singly linked list, given a pointer to the node before the one to be deleted.

Q8. [5 marks] (Medium)

Explain the concept of a Circular Queue and its primary benefit over a linear queue implemented with an array.

Q10. [5 marks] (Hard)

Discuss a practical application of a queue where its FIFO (First-In, First-Out) property is crucial for correct operation.

--- End of Question Paper ---